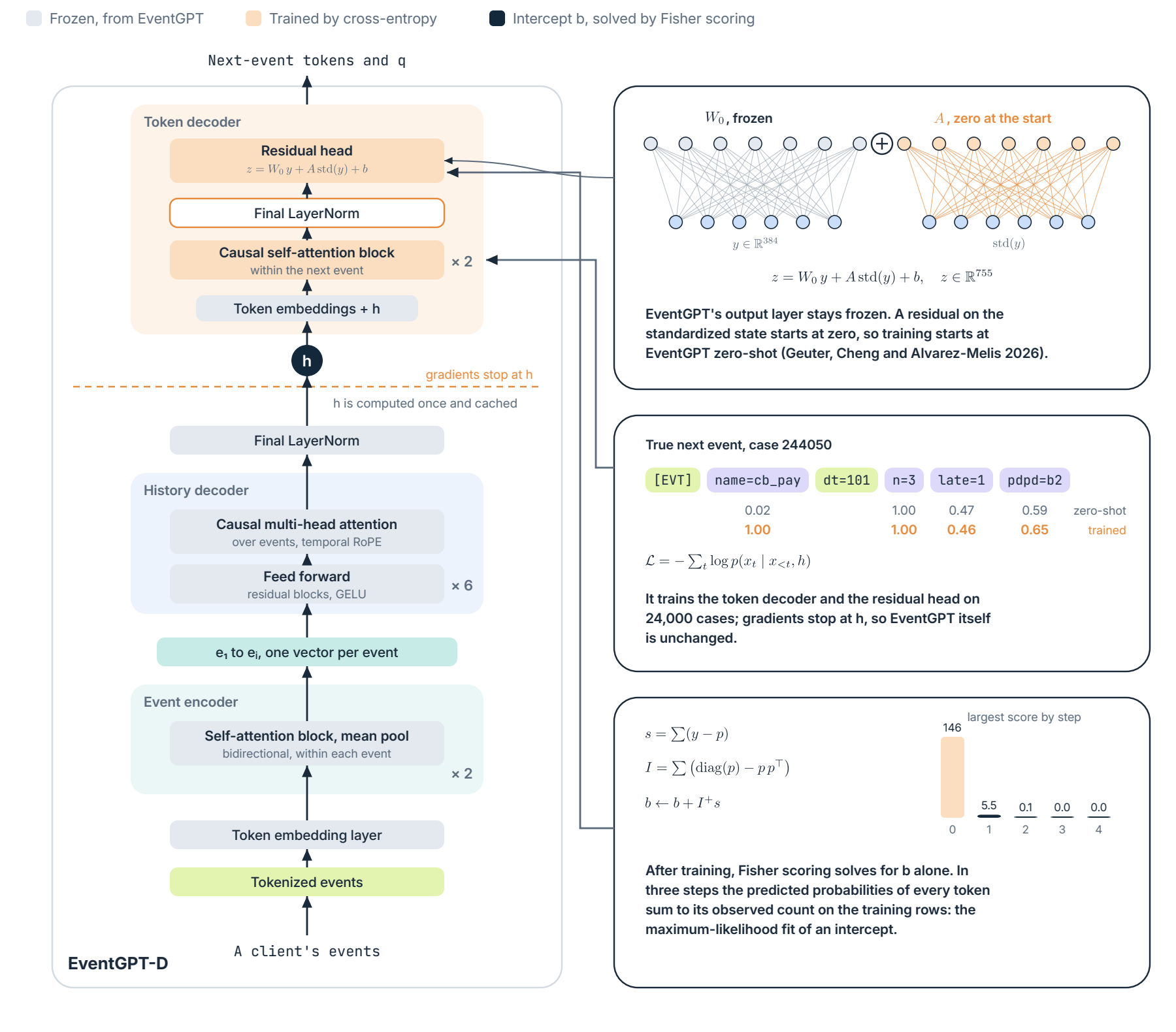

Frozen, from EventGPT
Trained by cross-entropy
Intercept b, solved by Fisher scoring
Next-event tokens and q
Token decoder
Residual head
$z = W_0\, y + A\,\mathrm{std}(y) + b$
Final LayerNorm
Causal self-attention block
within the next event
× 2
Token embeddings + h
h
gradients stop at h
h is computed once and cached
Final LayerNorm
History decoder
Causal multi-head attention
over events, temporal RoPE
Feed forward
residual blocks, GELU
× 6
e₁ to eᵢ, one vector per event
Event encoder
Self-attention block, mean pool
bidirectional, within each event
× 2
Token embedding layer
Tokenized events
A client's events
EventGPT-D
W_0, frozen
A, zero at the start
$y \in \mathbb{R}^{384}$
$\mathrm{std}(y)$
$z = W_0\, y + A\,\mathrm{std}(y) + b, \quad z \in \mathbb{R}^{755}$
EventGPT's output layer stays frozen. A residual on the standardized state starts at zero, so training starts at EventGPT zero-shot (Geuter, Cheng and Alvarez-Melis 2026).
True next event, case 244050
[EVT]
name=cb_pay
dt=101
n=3
late=1
pdpd=b2
0.02
1.00
1.00
1.00
0.47
0.46
0.59
0.65
zero-shot
trained
$\mathcal{L} = -\sum_t \log p(x_t \mid x_{<t}, h)$
It trains the token decoder and the residual head on 24,000 cases; gradients stop at h, so EventGPT itself is unchanged.
$s = \sum (y - p)$
$I = \sum \left(\mathrm{diag}(p) - p\,p^\top\right)$
$b \leftarrow b + I^{+} s$
largest score by step
146
5.5
0.1
0.0
0.0
0
1
2
3
4
After training, Fisher scoring solves for b alone. In three steps the predicted probabilities of every token sum to its observed count on the training rows: the maximum-likelihood fit of an intercept.